

TARGET-SPECIFIC RECOVERABILITY MAPS FOR REDUCED-LEAD ECG RECONSTRUCTION

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ABSTRACT

Reconstructing missing electrocardiogram (ECG) leads from a reduced set is ill-posed, and a scalar reconstruction error says nothing about *which* named feature’s low-rank (dipolar) component, on *which* lead, is even recoverable. We give a *target-specific recoverability map*: for a waveform segment s (P/QRS/ST/T), an observed lead set S , and a target lead ℓ , two closed-form numbers derived from one truncated SVD of the segment’s empirical rank-3 spatial (“dipolar”) sub-matrix—an *identifiability* $\eta_{s,\ell}(S)$ (zero iff the target’s low-rank component is recoverable from S) and a *conditioning* $\kappa_{s,\ell}(S)$ (its noise/observed-residual gain). The per-lead criterion is the classical estimability condition specialized to ECG (exact vs. a numerical ϱ -truncation, which we separate). The robust findings: precordial ST is *exactly* non-identifiable from limb leads (the limb leads span the frontal plane, so $M_{s,S}$ has exact rank 2), and the graded normalized-identifiability ordering is bootstrap-stable for QRS. For the low-amplitude ST/T segments the fine ordering is *sensitive to the limb-lead weighting* (a duplicated-limb sensitivity analysis), so we report it with that caveat rather than as an exact graded claim. We attach distribution-free calibrated intervals for the empirically predictable residual (conformalized quantile regression, strict fold discipline, honestly stated within-group coverage), and observe that the total ST-threshold error rate is empirically similar across the evaluated reconstructors (not a certified bound) while the false-positive/false-negative split is a reconstructor choice. The map turns “trust the reconstruction” into a per-feature, per-lead, calibrated statement.

Index Terms— electrocardiography, inverse problems, identifiability, conformal prediction, trustworthy machine learning

1. INTRODUCTION

Wearable and reduced-lead ECG acquisition makes lead *reconstruction* ubiquitous: recover the standard 12 leads from one, three, or six observed leads [1, 2]. A low aggregate error certifies nothing about *which* morphological features are trustworthy: a cardiologist reads the P wave, the QRS complex, the ST segment and the T wave, each on specific leads.

We ask a sharper, reconstructor-independent question: *for*

a given observed lead set, which target lead’s dipolar morphology is even identifiable, and how well-conditioned is its recovery? Each waveform segment’s instantaneous potential is approximately low rank across leads; we estimate a per-segment rank-3 spatial basis M_s (top-3 singular vectors of the population lead covariance). We stress that M_s is a *population-estimated (PCA)* object: on real PTB-XL its three principal angles to the classical vectorcardiographic (inverse-Dower) space are graded—one close ($2\text{--}6^\circ$), one moderate ($10\text{--}15^\circ$), and one substantially different ($43\text{--}55^\circ$)—so we call it an empirical rank-3 subspace rather than a physical dipole (Sec. 4).

Related work and contributions. Classical work reconstructs missing leads (regression / VCG synthesis [3, 4, 5], deep and arbitrary-mask waveform-completion models [1, 2, 6, 7]) or *selects* optimal leads [8]; whether a given target is even recoverable is the classical estimability question [9]. Those systems produce a reconstruction; we instead certify, before any reconstruction, which target is recoverable and attach a graded, per-target-lead recoverability measure with calibrated per-record error. Our neural baseline is a *representative* masked-completion U-Net (same family as [1, 6]), not a state-of-the-art claim. We do not claim the underlying identity as new. Our contributions are: (1) a *graded*, per-target-lead recoverability map—continuous $\eta_{s,\ell}(S)/\kappa_{s,\ell}(S)$ from a single truncated SVD that *rank* target leads (not a binary rank test), over an *empirically-estimated* M_s that differs $43\text{--}55^\circ$ from the textbook inverse-Dower dipole, with a truncation-tolerance sensitivity analysis and record-bootstrap CIs (Sec. 2); (2) a calibration layer attaching distribution-free intervals (a sound application of conformalized quantile regression + Mondrian [10], not a new method) to the map per feature and lead, with strict fold discipline and honest within-group (not per-diagnosis) coverage (Sec. 3); (3) a graded ST-safety study on PTB-XL—continuous full-waveform reconstruction shows anterior precordial ST is far from recoverable from limb leads, with a total ST-threshold error rate empirically similar across the evaluated reconstructors (an empirical observation, not a certified bound) and a false-positive/false-negative split that is a reconstructor choice (Sec. 4).

2. THE RECOVERABILITY CERTIFICATE

Model. Within segment s write the population 12-lead vector as $\mathbf{L}_s = \boldsymbol{\mu}_s + \mathbf{M}_s \mathbf{d} + \mathbf{r}_s$, where $\mathbf{M}_s \in \mathbb{R}^{12 \times 3}$ is the (population-estimated, PCA) dipolar basis, $\mathbf{d}$ the dipole coordinates, and $\mathbf{r}_s$ the non-dipolar residual. Observing S gives $\mathbf{y}_S = \text{Sel}_S \mathbf{L}_s + \mathbf{n}$, with $\mathbf{M}_{s,S} = \text{Sel}_S \mathbf{M}_s$. Write the exact Moore–Penrose pseudo-inverse $\mathbf{M}_{s,S}^+$ and the exact row-space projector $\mathbf{P}_{\text{obs}} = \mathbf{M}_{s,S}^+ \mathbf{M}_{s,S}$. For numerical deployment we also use the *rank- r truncation* at relative tolerance ϱ , $\mathbf{M}_{s,S}^{(\varrho)} = \mathbf{U}_r \boldsymbol{\Sigma}_r \mathbf{V}_r^\top$ keeping singular triplets with $\sigma_i > \varrho \sigma_{\max}$, with pseudo-inverse $\mathbf{M}_{s,S}^{(\varrho)+}$ and projector $\mathbf{P}_{\text{obs}}^{(\varrho)} = \mathbf{V}_r \mathbf{V}_r^\top$.

Proposition 1 (Exact per-target-lead dipolar identifiability and conditioning). *For each target lead ℓ define $\eta_{s,\ell}(S) = \|\mathbf{e}_\ell^\top \mathbf{M}_s (\mathbf{I} - \mathbf{P}_{\text{obs}})\|_2$ and $\kappa_{s,\ell}(S) = \|\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{M}_{s,S}^+\|_2$ from the exact $\mathbf{M}_{s,S}$.*

(i) *Identifiability. $\eta_{s,\ell}(S) = 0$ iff $\mathbf{e}_\ell^\top \mathbf{M}_s$ lies in the exact row space of $\mathbf{M}_{s,S}$; then $\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{d}$ is a deterministic function of $\mathbf{M}_{s,S} \mathbf{d}$ —two dipoles with the same observation have the same lead- ℓ dipolar value, at any SNR. If $\eta_{s,\ell}(S) > 0$ there is a $\boldsymbol{\delta} \in \ker \mathbf{M}_{s,S}$ (an unobserved dipole direction) with $\mathbf{e}_\ell^\top \mathbf{M}_s \boldsymbol{\delta} \neq 0$; then $\mathbf{d}$ and $\mathbf{d} + t\boldsymbol{\delta}$ produce identical observations for all t while lead ℓ ’s dipolar value changes, so it is not identifiable at any SNR.*

(ii) *Conditioning. On the identifiable part, the lead- ℓ error from observation noise and observed non-dipolar residual obeys $|\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{M}_{s,S}^+ (\text{Sel}_S \mathbf{r}_s + \mathbf{n})| \leq \kappa_{s,\ell}(S) (\|\text{Sel}_S \mathbf{r}_s\|_2 + \|\mathbf{n}\|_2)$.*

The global $\kappa_s(S) = \|\mathbf{M}_s \mathbf{M}_{s,S}^+\|_2$ (spectral norm) upper-bounds every per-lead $\kappa_{s,\ell}(S)$ (a row norm never exceeds the spectral norm; it is not their maximum).

Exact vs. ϱ -identifiability (numerical deployment). The deployed map computes $\eta_{s,\ell}^{(\varrho)}, \kappa_{s,\ell}^{(\varrho)}$ from the truncation $\mathbf{P}_{\text{obs}}^{(\varrho)}$, so a singular direction observed only through a value below $\varrho \sigma_{\max}$ is treated as unobserved. Then $\eta_{s,\ell}^{(\varrho)} > 0$ means lead ℓ is *not stably recoverable at resolution ϱ* (outside the retained numerical row space); it does *not* by itself imply the exact matrix has a null direction—a tiny-but-nonzero singular value is exactly identifiable ($\eta = 0$) yet ϱ -unidentifiable ($\eta^{(\varrho)} > 0$), as we verify numerically (`tests/test_exact_vs_rho.py`). We therefore call the $\eta^{(\varrho)}$ heatmap the *ϱ -identifiability map*; for well-separated configurations the exact and ϱ verdicts coincide, and near-rank-deficient S are reported with a ϱ -sweep and record-bootstrap CIs. **The limb-6 flagship is exact:** the six limb leads are exact linear combinations of I and II (Einthoven/Goldberger), so $\mathbf{M}_{s,S}$ has exact rank 2 and precordial ST is *exactly* non-identifiable from limb leads at any SNR, independent of ϱ . This exactness is conditional on the

a-priori rank-3 dipole model for $\mathbf{M}_s$: at retained rank 2 the limb-6 observation spans all of $\mathbf{M}_s$ and the precordial verdict collapses to $\eta=0$; the non-identifiability appears at rank ≥ 3 (the third dipolar direction), so “exact” means exact *given* rank 3.

Relation to classical estimability. Part (i) is the classical estimability criterion (Gauss–Markov / [9]) for the linear functional $\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{d}$; we claim no novelty for this identity. The contribution (Sec. 4) is its use as a graded, empirically-estimated recoverability map with per-record calibrated error rates (Sec. 3).

The non-dipolar residual (no independence claim on real data). Let $\mathbf{m}_s(\mathbf{y}_S) = \mathbb{E}[\mathbf{r}_s | \mathbf{y}_S]$ and $\mathbf{u}_s = \mathbf{r}_s - \mathbf{m}_s(\mathbf{y}_S)$, so $\mathbb{E}[\mathbf{u}_s | \mathbf{y}_S] = \mathbf{0}$. We do *not* assume $\mathbf{u}_s \perp \mathbf{y}_S$. Under squared loss the Bayes-optimal reconstruction error of the (vector) residual is

$$\inf_{\hat{\mathbf{r}}} \mathbb{E} \|\mathbf{r}_s - \hat{\mathbf{r}}(\mathbf{y}_S)\|_2^2 = \mathbb{E} [\text{tr Cov}(\mathbf{r}_s | \mathbf{y}_S)].$$

Part of $\mathbf{r}_s$ (the *empirically predictable residual*) is recovered by a predictor trained on $\mathbf{y}_S$ and metered with distribution-free calibrated intervals; the remainder is an *achievability gap*, established by a held-out achievability analysis for a stated predictor family—not declared unrecoverable a priori.

Proposition 2 (Strict independence limit – synthetic model only). *If the data are generated so that a component $\mathbf{u}$ is statistically independent of $\mathbf{y}_S$ (an explicit synthetic construction), then $p(\mathbf{u} | \mathbf{y}_S) = p(\mathbf{u})$, the Bayes estimate of $\mathbf{u}$ under squared loss is its prior mean, and $\mathbb{E} \|\hat{\mathbf{u}} - \mathbf{u}\|_2^2 \geq \text{tr Cov}(\mathbf{u})$ for any estimator. This strict limit is invoked only for the synthetic model; on real ECG we make no such independence claim.*

3. DISTRIBUTION-FREE CALIBRATION

3.1. Calibrated intervals for the predictable residual

The prediction target is the *record-level segment-mean* off-dipole residual of the target lead (one scalar per record: the mean over the segment’s samples). We train a gradient-boosted quantile regressor of it on the observed leads’ segment means and apply Mondrian conformalized quantile regression [10] per (S, s, ℓ) group, with strict fold discipline: basis and model on folds 1–7, the regressor’s `max_iter` selected by pinball loss on fold 8, conformal calibration on fold 9, and a single evaluation on fold 10 (nominal $1-\alpha=0.90$). This is a sound *application* of established conformal machinery, not a new predictor; the contribution is attaching it, per feature and lead, to the recoverability map. Across the pre-registered (S, s, ℓ) groups, fold-10 within-group coverage clusters near nominal (0.87–0.94; e.g. {I,II,V1,V3,V5} QRS/V2 0.91, record-bootstrap CI [0.89, 0.93], mean width 0.150 mV).

What is and is not guaranteed. Mondrian CQR gives *within*-(S, s, ℓ)-group marginal coverage under exchangeability—not coverage conditional on diagnosis, which is not a conditioning variable. Diagnostic subgroups are therefore not guaranteed and the smallest ones fall below nominal: the worst is the HYP subgroup at coverage 0.72 ($n=85$), the expected finite-sample behaviour when conditioning on a variable the calibration does not stratify on. We report worst-subgroup coverage rather than hide it under the aggregate band, and we make no distribution-free claim beyond exchangeability with the calibration fold.

4. EXPERIMENTS

Data. PTB-XL [11] (21,799 twelve-lead records, 100/500 Hz, 10-fold split) with P/QRS/ST/T delineation by NeuroKit2 [12].

Reproducibility. Code, fixed and deterministic seeds, a pinned dependency manifest, result lineage (commit, a working-tree-clean flag, and dataset/split hashes in every JSON), and a one-command pipeline are released¹; every number in the tables is regenerated from result JSON (no hand-typed values). The neural baseline is a $24 \rightarrow 12$ -channel 1-D U-Net (base width 64, residual GroupNorm blocks, 60 epochs, Adam 2×10^{-4} , MSE on masked leads under random masks, 3 training seeds); the residual predictor is a gradient-boosted quantile regressor. The map is a deterministic function of the estimated $\mu_s, \bar{M}_s$ (record-level bootstrap CIs), not a held-out evaluation.

4.1. The recoverability map on PTB-XL

We estimate $\bar{M}_s$ (rank 3; the cardiac dipole is 3-D *a priori*, and the cumulative explained variance confirms rank 3 captures the bulk—a rank-sensitivity sweep $r=2 \dots 5$ is reported in `results/recoverability_maps.json`) per segment on the training NORM records, with RECORD-LEVEL bootstrap CIs (resample records, refit $\bar{M}_s$; $\varrho=10^{-2}$; Fig. 1). We plot the *normalized* identifiability $\tilde{\eta}_{s,\ell} = \eta_{s,\ell} / \|\bar{e}_\ell^\top \bar{M}_s\|_2$ —a dimensionless relative unobserved geometric gain (a ratio of L_2 norms, not a variance fraction). A single observed lead leaves every other lead unidentifiable; a dipole-spanning $\{I, II, V2\}$ or $\{I, II, V1, V3, V5\}$ makes all targets *exactly* identifiable ($\eta=0$).

The robust flagship is exact. The six limb leads are exact linear combinations of I and II, so $\bar{M}_{s,S}$ for limb-6 has *exact* rank 2: precordial ST is *exactly* non-identifiable from limb leads at any SNR, independent of ϱ (Prop. 1).

Graded ordering: stable for QRS, weighting-sensitive for ST/T. A duplicated-limb sensitivity analysis (`results/lead_ordering`) refit on the 8 algebraically independent leads, on the *same*

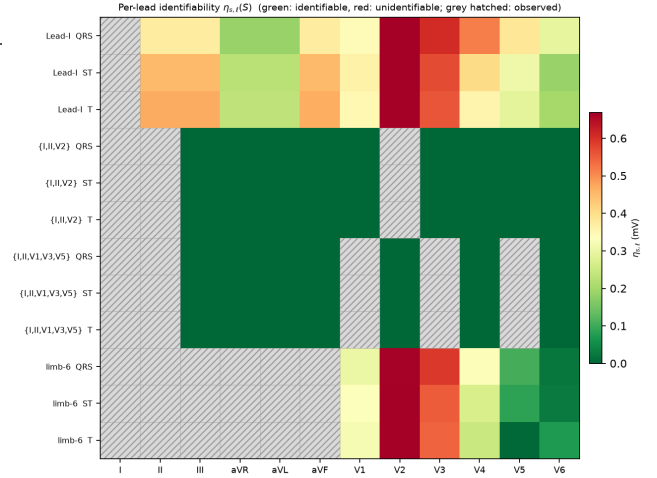


Fig. 1. Recoverability map: normalized identifiability $\tilde{\eta}_{s,\ell}(S)$ (left, dimensionless) and prior-conditional expected ambiguity in mV (right), colorblind-safe `cividis`; grey hatched = observed. For limb-6 all precordial leads are unidentifiable ($\eta>0$); the fine ST/T ordering is weighting-sensitive (text).

records) shows the V1–V6 $\tilde{\eta}$ ordering is bootstrap-stable for QRS (Spearman 1.00, CI lower bound 0.88; anterior-before-lateral in 0.98 of resamples) but *not* robust for the low-amplitude ST/T segments (ST Spearman point 0.89 with a wide bootstrap CI; anterior-before-lateral in only 0.77 of resamples). We therefore report the ST/T $\tilde{\eta}$ /ambiguity magnitudes with this caveat and *do not* claim a fine ST/T ordering; the binary limb→precordial verdict above is the robust ST statement. The estimated subspace also depends on diagnosis (a NORM vs. MI principal-angle sensitivity is reported), so the map is a *NORM-trained reference*.

Downstream ST cost across reconstructors. We reconstruct the full 10-second waveform with three real continuous linear reconstructors (observed leads exact; ST at J+60 ms vs. the isoelectric PR-segment baseline [P offset – QRS onset], on fiducials from the observed Lead II shared between truth and reconstruction; median over beats; the PR-segment baseline fell back to the pre-QRS window in $\approx 14\%$ of beats), and count ST-threshold events ($|ST| \geq 0.1$ mV; false positive = crossing in reconstruction not truth, false negative = vice versa; Table 1). On 1498 common records the *total* wrong-event rate is empirically similar across reconstructors (51.1–53.7%) and is *dominated by false negatives* (missed true crossings: limb→precordial recovery smooths ST toward the population), with the false-positive/false-negative split a reconstructor choice (paired record-bootstrap Δ significant). We do not derive a minimax/Bayes bound, so we call the shared total an empirically similar total error rate, *not* a certified lower bound; we report ST-threshold events, not diagnoses.

Truncation-tolerance sensitivity. Well-posed config-

¹Anonymized for review; released on acceptance. Camera-ready artifacts are regenerated from a clean tree (the lineage `git_dirty` flag is asserted false).

Table 1. Limb-6 $\rightarrow$ precordial (all V1–V6) ST reconstruction, 1498 common fold-10 records. FP/FN = false-positive/false-negative ST-threshold crossings (%); total = FP+FN. FN dominates (missed crossings); the split is a reconstructor choice. Auto-generated from `results/st_safety.json`.

reconstructor	FP %	FN %	total %	ST err (mV)
dipolar	5.3	45.7	51.1	0.077
ridge	1.7	52.1	53.7	0.076
OLS	1.7	51.8	53.5	0.076

urations are ϱ -stable ($\{I, II, V1, V3, V5\}$ is rank 3 across $\varrho \in \{10^{-4}, \dots, 10^{-1}\}$); near-rank ϱ -sensitive and reported with the ϱ -sweep and bootstrap CIs. A tiny-but-nonzero singular value is exactly identifiable yet ϱ -unidentifiable, which is why we separate exact from ϱ -identifiability (Prop. 1).

4.2. Baselines and what the map predicts

Table 2 reports reconstruction RMSE (mV) of the unobserved target leads under a *single, identical protocol* for all methods: the same shared split (train folds 1–7, ridge λ selected on fold 8, test on NORM fold-10), the same 800 test records, per-timepoint waveform RMSE on the same segment indices, and *paired* record-bootstrap CIs on the step-to-step differences. We include a *representative* masked-completion baseline—an arbitrary-mask conditional 1-D U-Net trained with random lead masks (one model serves any configuration, no target-lead leakage), the same family as [1, 6, 7], run over three seeds (mean reported)—not a state-of-the-art claim.

On a dipole-spanning set the methods form a ladder in point estimates (the ridge $\rightarrow$ U-Net step is within its paired CI)—prior-mean $\rightarrow$ dipolar (Tier I) $\rightarrow$ per-segment ridge (Tier I+II) $\rightarrow$ U-Net—evidence of recoverable non-dipolar content (the calibrated Tier II layer), though the neural model’s margin over a simple ridge is small (QRS 0.164 vs. 0.167 mV on $\{I, II, V1, V3, V5\}$; paired Δ CI $[-0.007, +0.000]$ mV). On **limb-6** the U-Net still *lowers* error over the prior mean (QRS 0.481 $\rightarrow$ 0.238)—it recovers the predictable, largely population-correlated structure—but plateaus at $1.46 \times$ its spanning-set error (paired spanning $\rightarrow$ limb-6 Δ CI $[+0.070, +0.079]$ mV) and no reconstructor closes that gap. This residual gap is *consistent* with the unobserved dipolar coordinate the map’s $\eta > 0$ marks as unidentifiable (we do not decompose the error into projected components, so we claim consistency, not equality): capacity recovers the predictable part but not the missing coordinate, so a reconstruction that looked confident about anterior precordial morphology here would be inventing it. We therefore scope the identifiability claim to the dipolar component, not to a blanket “not recoverable.”

Empirical, not physical, subspace. On real PTB-XL the

Table 2. Reconstruction RMSE (mV) of target leads, QRS segment, under one identical per-timepoint protocol on NORM fold-10 (800 test records, shared split). “dipolar” = Tier I; “ridge” = per-segment ridge (Tier I+II); “U-Net” = representative masked-completion baseline (3-seed mean). Best in bold. On limb-6 even the neural baseline plateaus far above its identifiable-set error. Numbers auto-generated from `results/fair_baselines.json`.

config	prior-mean	dipolar	ridge	U-Net
$\{I, II, V2\}$	0.467	0.236	0.220	0.183
$\{I, II, V1, V3, V5\}$	0.481	0.183	0.167	0.164
limb-6	0.481	0.361	0.344	0.238

estimated M_s has three graded principal angles to the classical inverse-Dower vectorcardiographic space [3, 4]: one close ($2\text{--}6^\circ$), one moderate ($10\text{--}15^\circ$), and one substantially different ($43\text{--}55^\circ$, bootstrap CIs excluding small angles). We therefore treat M_s as an *empirical rank-3 spatial subspace* estimated from the data, not a physical cardiac dipole; all recoverability quantities are stated with respect to this estimated subspace. This empirical, data-driven subspace—rather than a fixed physiological transform—is what makes the map dataset-specific and, we verify, stable across PTB-XL and Chapman for the QRS segment (and robust to the duplicated-limb weighting).

5. CONCLUSION

Reduced-lead ECG reconstruction has a *known low-rank structure per segment*. Exploiting it gives a graded per-target-lead recoverability map—what is identifiable, how well-conditioned, and (via calibration) what interval the predictable residual admits—before and independent of any reconstructor. It states which lead configurations support recovery of a feature’s low-rank component, e.g. that precordial ST is exactly non-identifiable from limb leads, and reports the empirically similar total ST-threshold error rate the evaluated reconstructors incur there (dominated by missed, not fabricated, crossings).

6. REFERENCES

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